

COMPLETE STRUCTURED CURRICULUM

Machine Learning Mastery Roadmap

From NumPy & Pandas foundations to production-grade ML —
a structured 6-phase journey inspired by CampusX 100 Days of ML.

6	~5	100+	12+
PHASES	MONTHS	TOPICS	PROJECTS

YOU ALREADY KNOW

Python Basics	NumPy	Pandas	High School Math
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01 PHASE 1 · WEEKS 1–3 ML Foundations & Mindset

~3 weeks · Building the mental model

WEEK 01

What is Machine Learning?

- AI vs ML vs Deep Learning
- Types of ML: Supervised, Unsupervised, Reinf
- Batch vs Online (Incremental) learning
- Instance-based vs Model-based learning
- Challenges in ML: overfitting, underfitting
- Real-world ML applications

PROJECT

Identify 5 ML use cases in apps you use daily

WEEK 02

ML Development Lifecycle

- MLDLC: problem framing to deployment
- How to frame an ML problem
- Data roles: Engineer vs Analyst vs Scientist
- Tensors: the building block of ML data
- Setup: Anaconda, Jupyter, Google Colab

PROJECT

Build a first end-to-end toy ML project

WEEK 03

Math Foundations for ML

- Vectors, Matrices & Multiplication
- Eigenvalues & Eigenvectors (intuition)
- Derivatives: maxima, minima, chain rule
- Probability: conditional probability, Bayes' theorem
- Distributions: Normal, Binomial, Bernoulli, Poisson

PROJECT

Implement matrix ops & probability problems in NumPy

PHASE 2 · WEEKS 4–7

02 Data Mastery — Getting & Understanding Data

~4 weeks · The most underrated skill in ML

WEEK 04

Data Acquisition

- Working with CSV files efficiently
- Reading JSON & SQL databases
- Fetching data from REST APIs
- Web scraping with BeautifulSoup
- Kaggle datasets & Google Dataset Search

PROJECT

Build a data pipeline: scrape + API + CSV → single

WEEK 05

Exploratory Data Analysis (EDA)

- Understanding data: dtypes, shape, describe()
- Univariate analysis: histograms, box plots
- Bivariate analysis: scatter, correlations
- Multivariate analysis: pair plots
- Automated EDA with YData Profiling

PROJECT

Full EDA on Titanic or IPL dataset with visual storytelling

WEEK 06

Statistics for ML

- Covariance & Pearson Correlation
- QQ Plots — testing normality
- Confidence Intervals & Hypothesis Testing
- Chi-square Test & ANOVA
- Central Limit Theorem (intuition)

WEEK 07

Data Visualization Mastery

- Matplotlib: figures, axes, subplots
- Seaborn: statistical plots
- Plotly: interactive charts
- Choosing the right chart for your data
- Storytelling with data

PROJECT

Create a visual EDA report for a business dataset

PHASE 3 · WEEKS 8–13

03 Feature Engineering & Preprocessing

~6 weeks · Where models are won or lost

WEEK 08

Feature Scaling

- Why scaling matters (and when it doesn't)
- Standardization (Z-score) — StandardScaler
- Normalization (Min-Max) — MinMaxScaler
- Robust Scaler for outlier-heavy data
- MaxAbs Scaler for sparse data

WEEK 09

Encoding Categorical Data

- Ordinal Encoding & Label Encoding
- One-Hot Encoding (dummy variable trap)
- Target Encoding & Frequency Encoding
- Binary Encoding for high-cardinality
- ColumnTransformer — per-column transforms

WEEK 10

Feature Transformation

- Log, Reciprocal, Square Root transforms
- Box-Cox & Yeo-Johnson Power Transforms
- Binning & Binarization / Discretization
- Handling mixed variable types
- Date & Time feature engineering

WEEK 11

Handling Missing Data

- MCAR, MAR, MNAR — types of missingness
- Complete Case Analysis
- Simple Imputation: mean, median, mode
- KNN Imputer — multivariate imputation
- MICE / Iterative Imputer
- Missing Indicator feature

WEEK 12

Outlier Detection & Removal

- Z-Score method
- IQR (Interquartile Range) method
- Percentile capping / Winsorization
- Isolation Forest (ML-based detection)
- When to remove vs cap vs keep outliers

WEEK 13

Feature Selection & Pipelines

- Filter methods: correlation, chi-square
- Wrapper methods: RFE, Forward/Backward
- Embedded: L1 regularization, feature importances
- sklearn Pipelines — chaining transforms
- FunctionTransformer — custom transforms

PROJECT

Full preprocessing pipeline on a real messy dataset

04 Core Supervised Learning Algorithms

~9 weeks · The heart of classical ML

WEEK 14

Linear Regression Deep Dive

- Simple & Multiple Linear Regression from scratch
- OLS — Ordinary Least Squares math
- Metrics: MAE, MSE, RMSE, R2, Adjusted R2
- Assumptions of Linear Regression
- Polynomial Regression

WEEK 15

Gradient Descent & Regularization

- Gradient Descent from scratch
- Batch vs Stochastic vs Mini-batch GD
- Learning rate tuning
- Ridge (L2) Regression
- Lasso (L1) Regression
- ElasticNet

PROJECT

Predict IPL player salaries or student exam scores

WEEK 16

Logistic Regression

- Sigmoid function — linear to probability
- Binary & Multi-class classification
- Accuracy, Precision, Recall, F1 score
- Confusion Matrix & ROC-AUC Curve
- Class imbalance: SMOTE, class_weight

PROJECT

Placement prediction: will a student get placed?

WEEK 17

Naive Bayes

- Bayes' Theorem applied to classification
- Gaussian, Multinomial, Bernoulli NB
- The conditional independence assumption
- Text classification: spam detection
- Laplace Smoothing

WEEK 18

K-Nearest Neighbors (KNN)

- Distance metrics: Euclidean, Manhattan
- Choosing K — elbow method
- KNN for regression vs classification
- Curse of dimensionality
- KD-Tree & Ball-Tree for faster KNN

WEEK 19

Support Vector Machines

- Maximal Margin Classifier intuition
- Soft-margin SVM & the C parameter
- Kernel Trick: RBF, Polynomial, Sigmoid
- SVM for regression (SVR)
- When to use SVM vs others

PROJECT

Digit recognition with SVM on MNIST

WEEK 20

Decision Trees

- How trees split: Gini vs Information Gain
- Pre-pruning vs Post-pruning
- Decision Trees for regression
- Visualizing and interpreting trees
- Why single trees overfit

WEEKS 21–22

Model Evaluation & Selection

- Bias-Variance Tradeoff
- Cross-Validation: k-Fold, Stratified k-Fold
- GridSearchCV, RandomizedSearchCV
- Optuna for Bayesian hyperparameter search
- Learning curves — diagnosing over/underfitting

PROJECT

Classifier with full evaluation report

05 Ensembles, Dimensionality Reduction & Unsupervised Learning

~10 weeks · Where ML gets powerful

WEEK 23

Ensemble: Bagging

- Bagging: Bootstrap Aggregating intuition
- Random Forest — bagging + feature randomness
- Feature importances from Random Forest
- OOB (Out-of-Bag) error estimation
- Extra Trees (Extremely Randomized Trees)

WEEK 24

Ensemble: Boosting

- AdaBoost — adaptive weighting
- Gradient Boosting Machines (GBM)
- XGBoost — why it wins Kaggle
- LightGBM — faster leaf-wise growth
- CatBoost — handles categoricals natively

PROJECT

Kaggle-style: predict with XGBoost + feature engineering

WEEK 25

Stacking & Blending

- Voting Classifiers (hard & soft voting)
- Stacking — model predictions as features
- Blending — a leak-free alternative
- When ensembling helps vs over-complicates

WEEKS 26–27

Dimensionality Reduction

- The Curse of Dimensionality
- PCA from scratch — eigenvalue decomposition
- Explained variance ratio & choosing components
- LDA — Linear Discriminant Analysis
- t-SNE & UMAP — visualization in 2D/3D

PROJECT

Visualize MNIST with t-SNE; speed up model with PCA

WEEKS 28–29

Clustering Algorithms

- K-Means from scratch
- Elbow method & Silhouette score
- DBSCAN — density-based clustering
- Hierarchical Clustering & Dendrograms
- Gaussian Mixture Models (GMM)

PROJECT

Customer segmentation for e-commerce dataset

WEEK 30

Anomaly Detection

- Isolation Forest
- One-Class SVM
- Local Outlier Factor (LOF)
- Autoencoders for anomaly detection (preview)
- Use cases: fraud, manufacturing QA

WEEK 31

Recommendation Systems

- Collaborative Filtering (user & item-based)
- Content-Based Filtering
- Matrix Factorization (SVD)
- Hybrid approaches
- Cold start problem & solutions

PROJECT

Movie recommender with MovieLens dataset

WEEK 32

Time Series Fundamentals

- Time series components: trend, seasonality
- Stationarity & ADF test
- ARIMA, SARIMA models
- Feature engineering for time series
- Using XGBoost on time series

PROJECT

Forecast stock prices or air passenger counts

06 Production ML & What's Next

~8 weeks · From notebook to real world

WEEK 33

Model Interpretability (XAI)

- Why interpretability matters
- SHAP values — unified feature attribution
- LIME — local surrogate models
- Permutation feature importance
- Partial Dependence Plots (PDP)

WEEKS 34–35

MLOps & Model Deployment

- Saving models: pickle, joblib, ONNX
- Building REST APIs with FastAPI
- Dockerizing your ML model
- Cloud deployment: AWS SageMaker / GCP intro
- MLflow for experiment tracking

PROJECT

Deploy a trained model as a FastAPI endpoint with Docker

WEEK 36

NLP Intro

- Text preprocessing: tokenization, stemming
- Bag of Words & TF-IDF
- Word embeddings: Word2Vec, GloVe (concept)
- Text classification with ML
- Introduction to Transformers & BERT

WEEK 37

Computer Vision Intro

- Image as a matrix — pixels, channels
- Feature extraction: HOG, SIFT concepts
- Image classification with traditional ML
- Introduction to CNNs (bridge to DL)

PROJECT

Sentiment analysis on IMDB reviews

WEEKS 38–39

Capstone Projects

- Project 1: complete ML pipeline of your choice
- Project 2: Kaggle competition submission (top)
- Portfolio: GitHub + write-ups + blog posts
- Code quality: clean notebooks, modular scripts

PROJECT

2 portfolio-ready projects published on GitHub/Medium

WEEK 40

What's Next — Deep Learning Bridge

- When ML is enough vs when you need DL
- Neural Networks: the next level
- Roadmap: CNNs → RNNs → Transformers → LLMs
- Resources: fast.ai, d2l.ai, Andrej Karpathy
- Interview prep: ML questions & system design

BASED ON CAMPUSX · 100 DAYS OF MACHINE LEARNING PLAYLIST · ENHANCED WITH INDUSTRY-STANDARD TOPICS